



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

REVIEW TEST

30-10-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & LIVE-I(A)]

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C,D	B,D	A,C	A,B,D	A,B	A,B,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	D	A	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	2.00	0.62	0.10	6.28	5.15 to 5.17	50.00	1.24	16.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C,D	A,B,C,D	A,B,C,D	A,B,C,D	A,C,D	C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	A	C	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	28.00	9.50	4.00	0.25	8.00	4.00	6.00	5.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C	B,C	A,B,C,D	C,D	A,C	A,B,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	A	D	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	9.00	72.00	6.00	1.00	9.00	10.00	5.00	7.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,B,C,D)

$$(A) K.E._{\max} = h\nu - \phi$$

$$2eV = \frac{6.63 \times 10^{-34} \times 9.5 \times 10^{15}}{2\pi \times 1.6 \times 10^{-19}} - \phi$$

$$2eV = 6.26 - \phi$$

$$\phi = 4.26 \text{ eV}$$

$$(B) P = \left[\frac{1}{2} \epsilon_0 E_0^2 C \right] A \sin 30^\circ$$

$$= \frac{1}{2} \times 8.85 \times 10^{-12} \times 225 \times 3 \times 10^8 \times 66.3 \times 10^{-4} \times \frac{1}{2} \omega$$

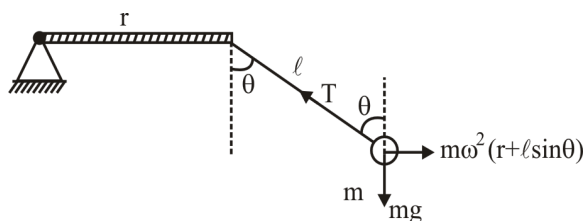
$$= 9.9 \times 10^{-4} \omega$$

$$(C) \eta = \frac{h\nu I}{pe} = \frac{6.26 \times 6 \times 10^{-6}}{9.9 \times 10^{-4}} = 0.038$$

$$(D) \frac{I_2}{I_1} = \left(\frac{E_2}{E_1} \right)^2$$

$$I_2 = \left(\frac{25}{15} \right)^2 (6\mu A) = 16.67\mu A$$

3. Ans (A,C)



$$\tan \theta = \frac{T \sin \theta}{T \cos \theta} = \frac{m\omega^2 (r + \ell \sin \theta)}{mg}$$

$$g = \frac{\omega^2 (r + \ell \sin \theta)}{\tan \theta}$$

$$g_m = \frac{(7)^2 \left[0.03 + 0.2 \times \frac{3}{5} \right]}{\left(\frac{3}{4} \right)} = 9.8 \text{ m/s}^2$$

$$g = \omega^2 \left(\frac{r}{\tan \theta} + \ell \cos \theta \right)$$

$$dg = \frac{\partial g}{\partial \omega} d\omega + \frac{\partial g}{\partial t} d\ell + \frac{\partial g}{\partial \theta} d\theta$$

$$= \left(\frac{r}{\tan \theta} + \ell \cos \theta \right) (2\omega d\omega) + \omega^2 (0 + \cos \theta) d\ell$$

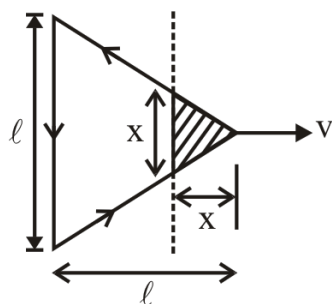
$$+ \omega^2 \left(-r \operatorname{cosec}^2 \theta - \ell \sin \theta \right) d\theta$$

$$\frac{dg}{g} = 2 \frac{d\omega}{\omega} + \frac{\cos \theta}{\left(\frac{r}{\tan \theta} + \ell \cos \theta \right)} d\ell$$

$$+ \left(\frac{r \operatorname{cosec}^2 \theta + \ell \cos \theta}{r \cot \theta + \ell \cos \theta} \right) d\theta$$

$$= 0.02 + 0.04 + 0.016 = 0.076$$

4. Ans (A,B,D)



$$\phi(x) = B \left(\frac{x^2}{2} \right) \cos 180^\circ$$

{ $\vec{A}$ is taken outside plane}

$$= -\frac{Bx^2}{2}$$

$$E(x) = -\frac{d\phi}{dt} = Bxv$$

$$I(x) = \frac{E(\phi)}{R} = \frac{Bxv}{R}$$

$$F(x) = IxB = \frac{B^2 x^2 v}{R} = -mv \frac{dv}{dx}$$

$$\Rightarrow \int_{v_0}^v dv = -\frac{B^2}{mR} \int_0^x x^2 dx$$

$$v = v_0 - \frac{B^2 x^3}{3mR} = v_0 - \frac{v_0 x^3}{3L^3} = v_0 \left(1 - \frac{x^3}{3L^3} \right)$$

$$I = \frac{Bv_0}{R} \left[x - \frac{x^4}{3L^3} \right] \frac{dI}{dx} = \frac{Bv_0}{R} \left[1 - \frac{4}{3} \frac{x^3}{L^3} \right] = 0$$

$$F = \frac{B^2 v_0}{R} \left[x^2 - \frac{x^5}{3L^3} \right] \frac{dF}{dx}$$

$$= \frac{B^2 v_0}{R} \left(2x - \frac{5x^4}{3L^3} \right) = \frac{2B_0^2 v_0 x}{R} \left[1 - \frac{5x^3}{6L^3} \right]$$

5. Ans (A,B)

$$(A) C_{eq} = C_1 + C_2 = \frac{\epsilon (\pi - \phi) R^2}{2d} + \frac{\epsilon_0 \phi R^2}{2d}$$

$$= \frac{R^2}{2d} \left[(\pi - \phi) \epsilon + \epsilon_0 \phi \right] = \frac{\pi R^2}{2d} \left[\epsilon - \frac{\pi}{\pi} (\epsilon - \epsilon_0) \right]$$

$$(B) dW_b + dW_{ext} = dU$$

$$V^2 dC + \tau d\phi = \frac{1}{2} V^2 dC$$

$$\tau = -\frac{V^2}{2} \frac{dC}{d\phi} = -\frac{V^2}{2} \left[\frac{\pi R^2}{2d} \left\{ 0 - \frac{(\epsilon - \epsilon_0)}{\pi} \right\} \right]$$

$$= +\frac{V^2 R^2}{4d} (\epsilon - \epsilon_0)$$

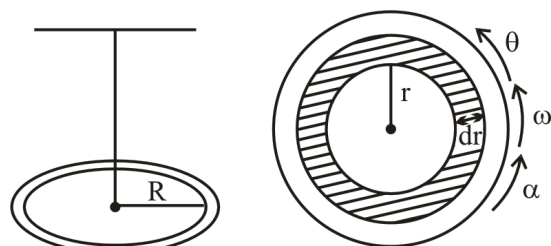
$$(C) dW_{ext} = dU$$

$$\tau d\phi = -\frac{Q^2}{2C^2} dC$$

$$\tau = -\frac{Q^2}{2C^2} \frac{dC}{d\phi} = -\frac{Q^2}{2C^2} \left[-\frac{R^2}{2d} (\epsilon - \epsilon_0) \right]$$

(D) as torque is constant motion will be oscillatory but not SHM.

6. Ans (A,B,D)



$$d\tau = r [(4\pi r dr) \eta v]$$

$$= 4\pi \eta r^2 dr (r\omega)$$

$$|dI_v = \int_0^R 4\pi \eta r^3 dr \omega$$

$$\tau_v = \pi \eta R^4 \omega$$

$$I\alpha = -C\theta - \tau_v = -C\theta - \pi \eta R^4 \omega$$

$$\frac{d^2\theta}{dt^2} + \frac{\pi \eta R^4}{I} \frac{d\theta}{dt} + \frac{C}{I} \theta = 0$$

$$\frac{d^2\theta}{dt^2} + 2\gamma \frac{d\theta}{dt} + \omega_0^2 \theta = 0$$

$$\omega_0 = \sqrt{\frac{C}{I}} = \sqrt{\frac{2C}{mR^2}} = 10 \text{ rad/s}$$

$$\gamma = \frac{\pi \eta R^4}{2I} + \frac{\pi \eta R^4}{2 \left(\frac{mR^2}{2} \right)} = \frac{\pi \eta R^2}{m}$$

For critically damped $\omega_0 = \gamma$

$$10 = \frac{\pi \eta \left(\frac{1}{4} \right)}{1}$$

$$\eta = \frac{40}{\pi} \text{ unit}$$

$$\text{For } \eta = \frac{24}{\pi} \quad \gamma = 6 \text{ rad/s}$$

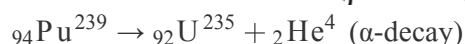
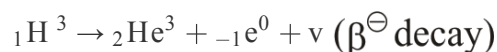
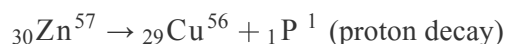
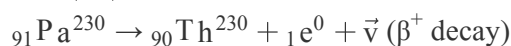
$$\omega_d = \sqrt{\omega_0^2 - \gamma^2} = \sqrt{100 - 36} = 8 \text{ rad/s}$$

$$Q\text{-factor} = \frac{\omega_0}{2\gamma} = \frac{10}{12} = \frac{5}{6}$$

PART-1 : PHYSICS

SECTION-I (ii)

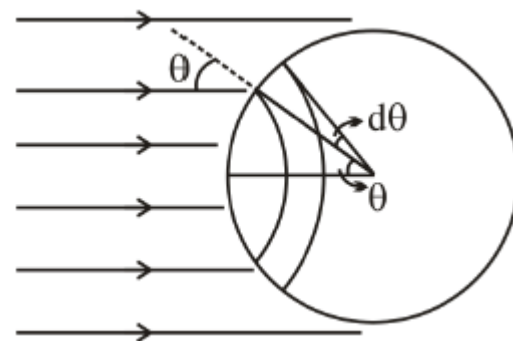
7. Ans (C)



PART-1 : PHYSICS

SECTION-II

1. Ans (2.00)



$$\text{Force due to absorption} = \frac{I (\pi R^2)}{2C}$$

Force due to reflection

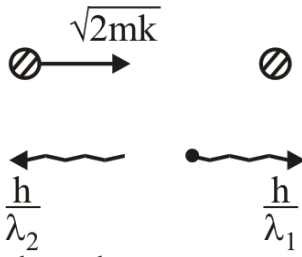
$$= \int_0^{\pi/2} \left(\frac{1}{C} \cos^2 \theta \right) [2\pi R \sin \theta] R d\theta \cos \theta$$

$$= 2\pi R^2 \left(\frac{1}{C} \right) \int_0^{\pi/2} \cos^3 \theta \sin \theta d\theta$$

$$= \frac{(\pi R^2) I}{2C}$$

$$\text{Total force on sphere} = \frac{I (\pi R^2)}{C}$$

2. Ans (0.62)



$$\frac{h}{\lambda_2} - \frac{h}{\lambda_1} = \sqrt{2mk}$$

$$\frac{hC}{\lambda_1} + \frac{hC}{\lambda_2} = 2mC^2 + k$$

$$\frac{hC}{\lambda_1} - \frac{hC}{\lambda_2} = \sqrt{2mk}C = \sqrt{2mC^2k}$$

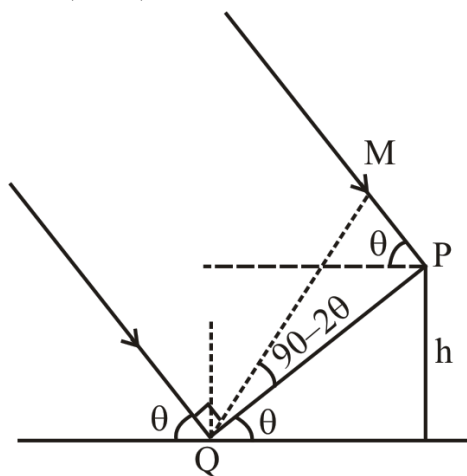
$$\frac{2hC}{\lambda_1} = 2mC^2 + k + \sqrt{2mC^2k}$$

$$\frac{hC}{\lambda_1} = mC^2 + \frac{k}{2} + \sqrt{\frac{mC^2k}{2}}$$

$$= 0.5\text{MeV} + 0.02\text{MeV} + \sqrt{\frac{0.5 \times 0.04}{2}}\text{MeV}$$

$$= 0.62\text{ MeV}$$

3. Ans (0.10)



$$\sin \theta = \frac{h}{PQ} \Rightarrow PQ = \frac{h}{\sin \theta}$$

$$PM = PQ \sin (90 - 2\theta)$$

$$= \frac{h}{\sin \theta} \cos 2\theta$$

$$\Delta x = \left(\frac{h}{\sin \theta} + \frac{\lambda}{2} \right) - \frac{h}{\sin \theta} \cos 2\theta$$

$$\Delta x = 2h \sin \theta + \frac{\lambda}{2}$$

For destructive interference

$$\Delta x = \frac{3\lambda}{2}$$

$$h = \frac{\lambda}{2 \sin \theta} = \frac{120\text{m}}{\sin 37} = 100\text{m}$$

$$= 0.1\text{ km}$$

5. Ans (5.15 to 5.17)

$$\ell = 40\text{ cm} \pm 0.1\text{ cm}$$

$$x = R \left(\frac{100 - \ell}{\ell} \right) = R \left[\frac{100}{\ell} - 1 \right]$$

$$dx = \frac{\partial x}{\partial R} dR + \frac{\partial x}{\partial \ell} d\ell$$

$$dx = \left(\frac{100}{\ell} - 1 \right) dR + R \left[-\frac{100}{\ell^2} d\ell \right]$$

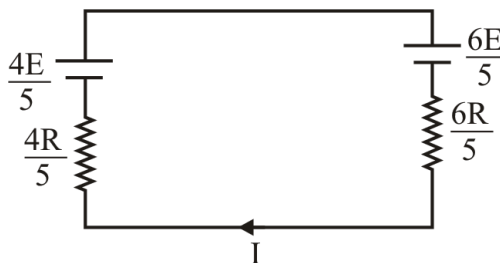
$$\frac{dx}{x} = \frac{dR}{R} + \frac{100}{\ell^2} \frac{d\ell}{\left(\frac{100}{\ell} - 1 \right)} = \frac{dR}{R} + \frac{100}{(100 - \ell)} \frac{d\ell}{\ell}$$

$$\frac{dx}{x} \times 100 = \left[\frac{0.1}{10} + \left(\frac{100}{100 - 40} \right) \left(\frac{0.1}{40} \right) \right] \times 100$$

$$= 1 + \frac{10}{6} \times \frac{10}{4} = 1 + 4.16$$

$$= 5.16\%$$

6. Ans (50.00)



$$I = \frac{2E/5}{10R/5} = \frac{E}{5R} = \frac{250\text{V}}{5 \times 1000\Omega} = 50\text{mA}$$

8. Ans (16.00)

$$T = 2\pi \sqrt{\frac{R^3}{GM_S}} = 1\text{yr} \dots (i)$$

$$2\pi \sqrt{\frac{(4R)^3}{G(4M)}} = 1\text{yr} \dots (ii)$$

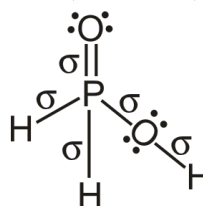
$$\frac{R^3}{GM_S} = \frac{16R^3}{GM}$$

$$\frac{M}{M_S} = 16$$

PART-2 : CHEMISTRY

SECTION-I (i)

2. Ans (A,B,C,D)



3. Ans (A,B,C,D)

$$\Delta G = \Delta H - T\Delta S$$

$$\Delta S < 0$$

$$\Delta G = \Delta H + T|\Delta S|$$

$$-nF E_{\text{Cell}} = \Delta H + T|\Delta S|$$

$$E_{\text{Cell}} = -\Delta H - T|\Delta S|$$

If temperature increases E_{Cell} decreases.

$$E^{\circ}_{\text{Cell}} > 0$$

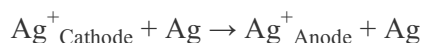
Because anode compartment has very small amount of concentration compare to cathode compartment due to PPT formation.

$$\Delta G^{\circ} = -nFE^{\circ}$$

$$\text{So, } \Delta G^{\circ} < 0$$

If anodic solution is diluted then Cl^{-} conc. decreases and Ag^{+} conc. increase as their product is constant.

$$[\text{Ag}^{+}][\text{Cl}^{-}] = K_{\text{SP}}$$



So, for cell emf to increase $\text{Ag}^{+}_{\text{Anode}}$ must decrease so since $\text{Ag}^{+}_{\text{Anode}}$ increasing the EMF will decrease.

Answer is ABCD.

4. Ans (A,B,C,D)

$$W_{\text{AB}} = -P(V_{\text{B}} - V_{\text{A}}) = -1(40 - 20) = -20 \text{ L-atm}$$

$$W_{\text{BC}} = 0$$

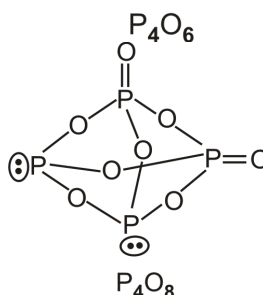
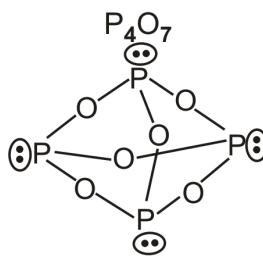
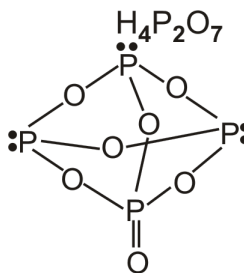
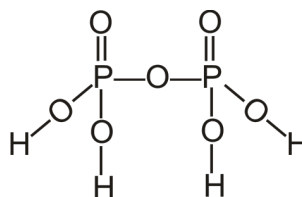
$$W_{\text{CA}} = -2.303 nRT \log \frac{V_{\text{A}}}{V_{\text{C}}} = -2.303 P_{\text{A}} V_{\text{A}} \log \frac{20}{40} = +13.86 \text{ L atm}$$

$$\Delta U = 0, \Delta H = 0, \Delta S = 0$$

PART-2 : CHEMISTRY

SECTION-I (ii)

7. Ans (A)



9. Ans (C)

$$(P) E_{\text{cell}} = -\frac{0.0591}{2} \log \frac{C}{2C} = 8.89 \times 10^{-3}$$

Spontaneous, cell works, conc. cell.

$$(Q) E_{\text{cell}} = -0.0591 \log \frac{1}{1} = 0$$

Non spontaneous, cell will not work conc. cell.

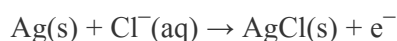
$$(R) E_{\text{cell}} = -\frac{0.0591}{2} \log \frac{0.01}{0.1^2} - 0.34 + 0.8 = 0.66$$

Spontaneous, cell works.

$$(S) E_{\text{cell}} = -\frac{0.0591}{1} \log \frac{\text{Cl}^{-}}{K_{\text{sp}}} - \frac{0.0591}{1} \log \frac{1}{[\text{Ag}^{+}]} = \frac{0.0591}{2} \log \frac{0.1 \times 0.1}{K_{\text{sp}}(\text{AgCl})} = +ve$$

Spontaneous, cell works, conc. cell.

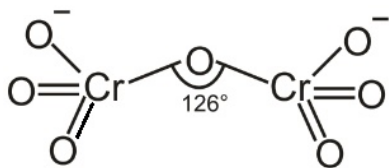
at anode,



PART-2 : CHEMISTRY

SECTION-II

1. Ans (28.00)



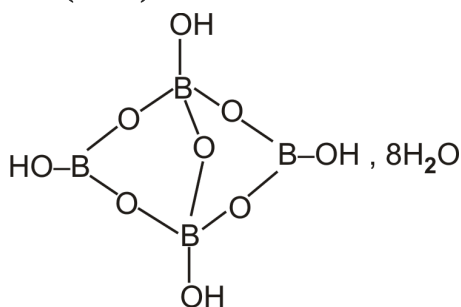
$$x = \sigma \text{ bond} = 8$$

$$y = \pi \text{ bond} = 4$$

$$a = \text{lone pair electron} = 16$$

$$8 + 4 + 16 = 28$$

2. Ans (9.50)



$$a = 34$$

$$b = 19$$

$$c = 34$$

$$d = 2$$

$$\frac{34}{34} \times \frac{19}{2} = 9.50$$

3. Ans (4.00)

$$\Lambda_{mHX} = \frac{\Lambda_{mHY}}{100}$$

$$[HX] = [HY] = 0.1 \text{ M} = C$$

$$\therefore \lambda_{x-}^0 = \lambda_{y-}^0$$

$$\Rightarrow \Lambda_{m,HX}^0 = \Lambda_{m,HY}^0$$

$$\frac{K_{aHX}}{K_{aHY}} = \frac{C\alpha_{HX}^2}{C\alpha_{HY}^2} = \left(\frac{\Lambda_{mHX}}{\Lambda_{mHY}} \right)^2 = \left(\frac{1}{100} \right)^2 = 10^{-4}$$

$$\text{or, } \log K_{aHX} - \log K_{aHY} = -4$$

$$\text{or, } pK_{aHX} + pK_{aHY} = -4$$

$$\text{or, } pK_{aHX} - pK_{aHY} = 4$$

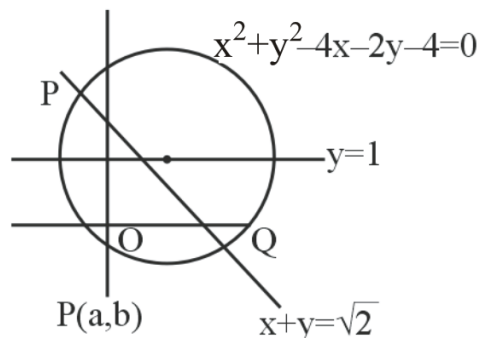
7. Ans (6.00)

(i), (iii), (iv), (v) (vi), (vii)

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,B,C)



$$OP = \sqrt{a^2 + b^2} = \sqrt{4a + 2b + 4} = \sqrt{2a + 2\sqrt{2} + 4}$$

$$|z| < 3$$

$$|\omega| < \sqrt{5} + \sqrt{3} < 4,$$

$$||z| - |\omega|| < |z - \omega| < |z| + |\omega|$$

$$|z - \omega| = \text{length of chord PQ} < \text{diameter}$$

$$||z| - |\omega|| < 6$$

$$-6 < |z| - |\omega| < 6$$

2. Ans (B,C)

D.R's of lines L_1, L_2, L_3 are same i.e. $\hat{i} - \hat{j} - \hat{k}$

so they are parallel.

$$\text{from cramer's rule } \Delta = \begin{vmatrix} 2 & 1 & 1 \\ 0 & 1 & -1 \\ 3 & 2 & 1 \end{vmatrix} = 0$$

$$\text{similarly } \Delta_1 = 0, \Delta_2 = 0, \Delta_3 = 0$$

infinite solution

3. Ans (A,B,C,D)

$$\frac{dy}{dx} - y \cot x = -\frac{\sin x}{x^2}$$

$$IF = e^{-\int \cot x} = \cos x$$

$$y \operatorname{cosec} x = \int -\frac{1}{x^2} dx$$

$$\frac{y}{\sin x} = \frac{1}{x} + c \Rightarrow y = \frac{\sin x}{x} + c \sin x$$

$$\lim_{x \rightarrow \infty} y = 0 \Rightarrow C = 0$$

$$f(x) = \frac{\sin x}{x}$$

$$\frac{\pi}{2} \cdot \frac{2}{\pi} < \int_0^{\pi/2} \frac{\sin x}{x} dx < \frac{\pi}{2}$$

$$1 < \int_0^{\pi/2} \frac{\sin x}{x} dx < \frac{\pi}{2}$$

4. Ans (C,D)

$P(E_1) = 1 - P(\text{unit's place in both is}$

1,2,3,4,6,7,8,9)

$$P(E_1) = 1 - \left(\frac{4}{5}\right)^2 = \frac{9}{25}$$

$P(E_2) = P(13579) - P(1379)$ for 2 numbers

$$= \frac{1}{4} - \frac{4}{25} = \frac{25-16}{100} = \frac{9}{100}$$

$$\frac{P(E_2)}{P(E_1)} = \frac{9}{100} \cdot \frac{9}{100} \cdot \frac{25}{9} = \frac{1}{4}$$

$P(E_1) = 4P(E_2) \Rightarrow A$ is not correct

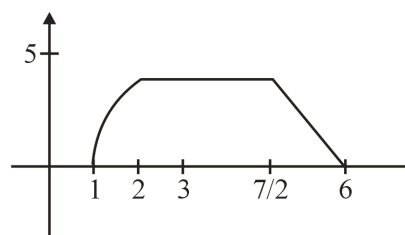
$$P(E_2/E_1) = \frac{P(E_2 \cap E_1)}{P(E_1)} = \frac{P(E_2)}{P(E_1)} = \frac{9}{100} \cdot \frac{25}{9} = \frac{1}{4} \Rightarrow (C)$$

$$P(E_1/E_2) = \frac{P(E_1 \cap E_2)}{P(E_2)} = \frac{P(E_2)}{P(E_2)} = 1 \Rightarrow (D)$$

5. Ans (A,C)

$$f(x) = 6x^2 - 30x + 36 = 6(x^2 - 5x + 6)$$

graph of $y = g(x)$



not diff. at $x = 7/2$ only

6. Ans (A,B,C)

$$f(|f(x) - f(y)|) = f(f(x)) - 2x^2 f(y) + f(y^2)$$

$$x = 0 = y \Rightarrow f(0) = f(f(0)) + f(0) \Rightarrow f(f(0)) = 0$$

$$x = 0 \Rightarrow f(|f(0) - f(y)|) = f(f(0)) - 0 + f(y^2)$$

$$f(|f(0) - f(y)|) = f(y^2)$$

$$|f(0) - f(y)| = y^2$$

$$f(y) = f(0) \pm y^2$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (C)

$$(I) A = B^{-1} \Rightarrow B = A^{-1}$$

$$|A| = 2 - 7 = -5$$

$$-x = -\frac{1}{5} (I) \Rightarrow x = \frac{1}{5}$$

$$(II) |A|^3 = (\alpha^2 - 4)^3 = 125 \Rightarrow \alpha = \pm 3$$

(III)

$$\Delta = \begin{vmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ac & bc & -c^2 \end{vmatrix} = a^2 b^2 c^2 \begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$

$$= 4a^2 b^2 c^2$$

(IV)

$$f(\theta) = -\sin \theta (-\cos^2 \theta \sin \theta - \sin^3 \theta) - \cos \theta (-\cos^3 \theta - \cos \theta \sin^3 \theta)$$

$$= \sin^2 \theta (1) + \cos^2 \theta (1) = 1$$

8. Ans (A)

(I) $(1, \alpha, 1)$ lies on $2x + y - 6z = 0$

$\alpha = 4$

and $2\beta - 6 + 12 = 0 \Rightarrow \beta = -3$

(II) $D = 0$

$$\Rightarrow \begin{vmatrix} 2 & -1 & 1 \\ 1 & -3 & 1 \\ \alpha & 1 & 1 \end{vmatrix} = 0 \Rightarrow 2(-4) + (1 - \alpha) + (1 + 3\alpha)$$

$\alpha = 3$

$$D_1 = \begin{vmatrix} 2 & -1 & 1 \\ \beta & -3 & 1 \\ 5 & 1 & 1 \end{vmatrix} = 0 \text{ for,}$$

$\beta = -1$ same for Δ_2, Δ_3 for no solution $\beta \neq -1$

$$(III) \begin{vmatrix} a & b & 1 \\ 1 & b & c \\ a & 1 & c \end{vmatrix} = 0 \Rightarrow 2abc - ab - bc - ca = -1$$

(IV) direction ratio of given line 1, 1, -2

equation of plane $x + y - 2z = -5$

9. Ans (D)

(I) $\frac{{}^{10}C_4 2^4}{{}^{20}C_4} = \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 2^4}{20 \cdot 19 \cdot 18 \cdot 17} = \frac{16 \times 14}{19 \times 17}$

(II) $1 - \frac{16 \times 14}{19 \times 19} = 1 - \frac{{}^{10}C_4 \cdot 2^4}{{}^{20}C_4}$

(III) $\frac{{}^{10}C_1 \times {}^9C_2 \cdot 2^2}{{}^{20}C_4} = \frac{10 \times 36 \times 4}{20 \cdot 19 \cdot 18 \cdot 17} \times 24 = \frac{96}{17 \times 19}$

(IV) $\frac{{}^{10}C_2}{{}^{20}C_4} = \frac{45 \times 24}{20 \cdot 19 \cdot 18 \cdot 17} = \frac{3}{19 \times 17}$

10. Ans (B)

(I)

$$I = \int_0^{\pi/4} \ln(\sin^2 2x) dx = 2 \int_0^{\pi/4} \ln(\sin 2x) dx$$

$$2x = t \Rightarrow dx = \frac{dt}{2}$$

$$I = \frac{2}{2} \int_0^{\pi/2} \ln(\sin t) dt = -\frac{\pi}{2} \ln 2$$

(II) $\int_0^{\pi/2} \underbrace{x}_{I} \underbrace{\sin 2x e^{\cos^2 x}}_{II} dx - \int e^{\cos^2 x} dx$

$$\left(-xe^{\cos^2 x} \right)_0^{\pi/2} + \int_0^{\pi/2} e^{\cos^2 x} dx - \int_0^{\pi/2} e^{\cos^2 x} dx$$

$$= -\frac{\pi}{2} e^0 = -\frac{\pi}{2}$$

(III)

$$f\left(x + \frac{\pi}{24}\right) + f\left(x + \frac{5\pi}{24}\right) = f\left(x + \frac{\pi}{8}\right)$$

$$f\left(x + \frac{3\pi}{24}\right) + f\left(x + \frac{7\pi}{24}\right) = f\left(x + \frac{5\pi}{24}\right)$$

$$f\left(x + \frac{\pi}{24}\right) + f\left(x + \frac{7\pi}{24}\right)$$

$$f\left(x + \frac{7\pi}{24}\right) + f\left(x + \frac{13\pi}{24}\right) = 0$$

$$f\left(x + \frac{\pi}{24}\right) = f\left(x + \frac{13\pi}{24}\right) \Rightarrow f(x) = f\left(x + \frac{\pi}{2}\right)$$

$$\int_{\pi/4}^{\pi/4 + \frac{3\pi}{2}} f(x) dx = 3 \Rightarrow 3 \int_0^{\pi/2} f(x) dx = 3$$

$$\int_0^{\pi/2} f(x) dx = 1 \Rightarrow \int_0^{\pi} f(x) dx = 2$$

(IV) $f(x) = \int \frac{\frac{5}{x^6} + \frac{4}{x^5}}{\left(1 + \frac{1}{x^4} + \frac{1}{x^5}\right)^2} dx$

$$= -\int \frac{dt}{t^2} = \frac{1}{t} + c$$

$$= \frac{x^5}{1 + x + x^5} + c$$

$c = 1$

$$f(-1) = -\frac{1}{1 - 1 - 1} + 1 = 2$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (9.00)**

$$I_n = \int_0^{\infty} x^{2n} e^{-x^2} dx$$

$$\left(e^{-x^2} \frac{x^{2n+1}}{2n+1} \right)_0^{\infty} - \int_0^{\infty} \frac{x^{2n+1}}{2n+1} (e^{-x^2}) (-2x) dx$$

$$= 0 + \frac{2}{2n+1} \int_0^{\infty} x^{2n+2} e^{-x^2} dx$$

$$I_n = \frac{2}{2n+1} I_{n+1}$$

$$\frac{I_{n+1}}{I_n} = \frac{2n+1}{2}$$

$$n=4 \Rightarrow \frac{I_5}{I_4} = \frac{9}{2}$$

$$n=3 \Rightarrow \frac{I_4}{I_3} = \frac{7}{2}$$

$$\frac{I_5}{I_3} = \frac{63}{4} \Rightarrow \frac{4I_5}{7I_3} = 9$$

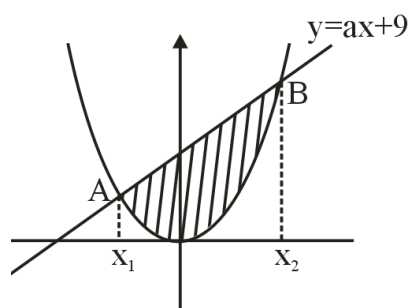
2. **Ans (72.00)**

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(f(x) + (h) + h\sqrt{f(x)}) - (f(x) + f(0))}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} + \sqrt{f(x)}$$

$$f'(x) = \sqrt{f(x)} \Rightarrow 2\sqrt{f(x)} = x + c$$



$$f(x) = \frac{(x+c)^2}{4} \Rightarrow f(x) = \frac{x^2}{4}$$

$$\int_{x_1}^{x_2} \left(ax + 9 - \frac{x^2}{4} \right) dx = A$$

$$A = \left(a \frac{x^2}{2} + 9x - \frac{x^3}{12} \right)_{x_1}^{x_2}$$

$$= \frac{a}{2} (x_2^2 - x_1^2) + 9(x_2 - x_1) - \frac{1}{12} (x_2^3 - x_1^3)$$

$$(x_2 - x_1) \left[\frac{a}{2} (x_2 + x_1) + 9 - \frac{1}{12} (x_1^2 + x_2^2 + x_1 x_2) \right]$$

$$x^2 - 4ax - 36 = 0 \begin{cases} x_1 \\ x_2 \end{cases}$$

$$= \sqrt{16a^2 + 144} \left[\frac{2a^2 + 18}{3} \right] = \frac{8}{3} (a^2 + 9)^{3/2}$$

minimum when $a = 0$

Area = 72 sq. unit

3. **Ans (6.00)**

$$\text{Let } a = p^{1/3}, b = p^{1/3} \omega, c = p^{1/3} \omega^2$$

$$\left| \frac{xa + yb + zc}{xb + yc + za} \right| = \left| \frac{x + y\omega + z\omega^2}{x\omega + y\omega^2 + z} \right| = \left| \frac{1}{\omega} \right| = 1$$

$$1 + (a_1^2 - 2b_1^2) \omega + a\omega^2 = 0 \text{ where } [x] + [y] + [z] = a$$

$$1 + (a_1^2 - 2b_1^2) \left(\frac{-1 + i\sqrt{3}}{2} \right) + a \left(\frac{-1 - i\sqrt{3}}{2} \right) = 0$$

$$\left(1 - \frac{1}{2} (a_1^2 - 2b_1^2) - \frac{a}{2} \right) + \frac{i\sqrt{3}}{2} (a_1^2 - 2b_1^2 - a) = 0$$

$$a_1^2 - 2b_1^2 + a = 2, a_1^2 - 2b_1^2 = a$$

$$a = 1$$

$$a_1^2 - 2b_1^2 = 1 \Rightarrow a_1^2 = 1 + 2b_1^2 \text{ (} a_1 \text{ is odd)}$$

$$(2n+1)^2 = 1 + 2b_1^2 \Rightarrow 4n^2 + 4n + 1 = 1 + 2b_1^2$$

$$b_1^2 = 2n(n+1) \text{ (} b \text{ is even)}$$

$$b_1 = 2, a_1 = 3$$

$$[x] + [y] + [z] + 2 + 3 = 6$$

4. **Ans (1.00)**

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{n}{r^2 - n^2 x^2} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \frac{1}{\frac{r^2}{n^2} - x^2}$$

$$= \int_0^1 \frac{dy}{y^2 - x^2} = \frac{1}{2x} \left[\ln \left| \frac{y-x}{y+x} \right| \right]_0^1$$

$$= \frac{1}{2x} \ln \left(\frac{1-x}{1+x} \right)$$

$$\left| \lim_{x \rightarrow 0.2} \frac{1}{2} \left(\frac{\ln(1-x)}{x} - \frac{\ln(1+x)}{x} \right) \right| = \left| \frac{1}{2} (-1-1) \right| = 1$$

5. Ans (9.00)

equation of lines AB and CD

$$\vec{r} = (9\hat{i} - \hat{j} + 7\hat{k}) + \lambda (4\hat{i} - \hat{j} + 3\hat{k})$$

$$\vec{r} = (7\hat{i} - 2\hat{j} + 7\hat{k}) + \gamma (2\hat{i} - \hat{j} + 2\hat{k})$$

point of intersection P(1,1,1)

$$\vec{PQ} \parallel \vec{AB} \times \vec{CD}$$

$$\vec{AB} \times \vec{CD} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -1 & 3 \\ 2 & -1 & 2 \end{vmatrix} = \hat{i} - 2\hat{j} - 2\hat{k}$$

$$\vec{PQ} = 5(\hat{i} - 2\hat{j} - 2\hat{k}) \text{ or } \vec{PQ} = -5(\hat{i} - 2\hat{j} - 2\hat{k})$$

$$\text{pv or Q} = 6\hat{i} - 9\hat{j} - \hat{k} \text{ or } -4\hat{i} + 9\hat{j} + 9\hat{k}$$

6. Ans (10.00)

$$z = i \left(\frac{-1 - i\sqrt{3}}{2} \right) = i\omega^2$$

$$\left((i\omega^2)^{101} + i^{109} \right)^{106} = (i\omega + i)^{106} = -(1 + \omega)^{106}$$

$$= -(-\omega^2)^{106} = -(\omega^2)$$

$$-\omega^2 = i^n \omega^{2n} \Rightarrow e^{i\pi/3} = (i\omega^2)^n$$

$$e^{i\pi/3} = \left(e^{-i\pi/6} \right)^n$$

$$-\frac{n\pi}{6} = \frac{\pi}{3} + 2k\pi$$

$$\text{for least positive } n \Rightarrow \frac{n\pi}{6} = -\frac{\pi}{3} + 2\pi \Rightarrow n = 10$$

7. Ans (5.00)

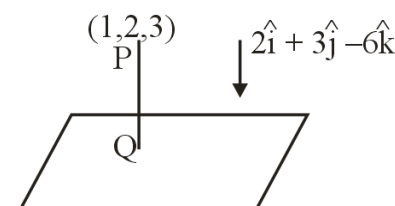
$$\text{No. of different ticks} = {}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4 = 15$$

$$\text{probability of getting correct ans} = \frac{1}{15}$$

Let he is given n number of chances

$$\frac{n}{15} \geq \frac{1}{3} \Rightarrow n \geq 5$$

8. Ans (7.00)



equation of PQ

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{-6}$$

Let Q(2λ + 1, 3λ + 2, -6λ + 3)

Q lies on x + y + z = 5

$$-\lambda + 6 = 5 \Rightarrow \lambda = 1$$

$$Q(3, 5, -3), PQ = \sqrt{4 + 9 + 36} = 7$$

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & LIVE-I(A)]

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	D	B	C				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	B,D	A,B,D	A,B,C	C,D	A,B,C	A,B,C,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	1	3	2	9	1	1	2	2

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	D	A	B	D				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,B,C,D	B,C	A,B,C	B,C	A,B,C,D	A,B,C,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	6	4	0	9	4	6	8	4

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	D	D	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,B,C,D	C	B,C	A,B,C,D	A,B,C	A,C		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	1	3	7	4	2	9	2	1

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

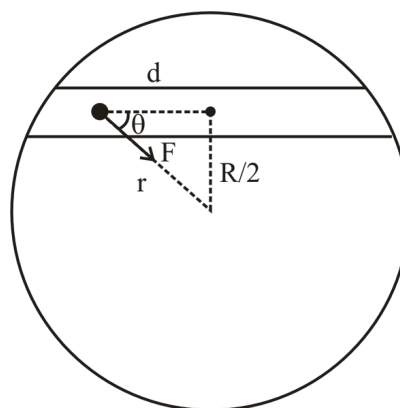
1. Ans (A)

$$f = \frac{C}{\lambda} = RC \left[\frac{1}{(n-m)^2} - \frac{1}{n^2} \right]$$

$$\frac{RC}{n^2} \left[\left(1 - \frac{m}{n} \right)^{-2} - 1 \right] = \left(\frac{2RC}{n^3} \right) m$$

For $m = 1, 2, 3, \dots$ f are in A.P.

2. Ans (D)



$$F = mg' = \frac{mgr}{R}$$

$$N = F \sin \theta$$

$$= \frac{mgr}{R} \times \frac{\frac{R}{2}}{r} = \frac{mg}{2}$$

3. **Ans (B)**

$$W = E - KE$$

$$= \frac{hc}{\lambda} - \frac{hc}{2\lambda} = \frac{hc}{2\lambda}$$

4. **Ans (C)**

$$C = \frac{\epsilon_0 A}{\ell}$$

$$\ln C = \ln \epsilon_0 + \ln A - \ln (\ell)$$

$$\frac{dC}{C} = 0 + \frac{dA}{A} - \frac{d\ell}{\ell}$$

$$0 = 2\alpha_1 - \alpha_2 \Rightarrow 2\alpha_1 = \alpha_2$$

PART-1 : PHYSICS

SECTION-I (ii)

5. **Ans (B,D)**

$$\text{Since } E \text{ will be defined as } E = \frac{kq}{r^2}$$

$$\text{So we can calculate } \oint_S \vec{E} \cdot d\vec{S} \text{ also.}$$

So only option (C) is incorrect.

6. **Ans (A,B,D)**

E on axis

$$E = \frac{kQx}{(R^2 + x^2)^{\frac{3}{2}}}$$

$\therefore$ For small x ($R \ll R$)

$$E = \frac{kQ}{R^3} x$$

$\therefore$ restoring force if $q < 0$

$$F = \frac{kQq}{R^3} x$$

$$a = - \left(\frac{kQq}{mR^3} \right) x$$

$\therefore$ compare with

$$a = -\omega^2 x$$

$$T = 2\pi \sqrt{\frac{mR^3}{kQq}}$$

7. **Ans (A,B,C)**

Convex lens (+ve power) can form enlarged & of same size real inverted image so A & C are correct.

Concave lens (-ve power) can form virtual diminished erect image of real object (B is correct).

8. **Ans (C,D)**

$$\vec{F}_e = q\vec{E} \text{ so equal \& opposite}$$

$$\vec{F}_m = q (\vec{v} \times \vec{B})$$

$\vec{F}_m$ is equal on both charge since q & v both are -ve for positron

W.D by magnetic force is zero so (C) is correct.

$\vec{F}_e$ is zero on system so (D) is correct.

9. **Ans (A,B,C)**

Formula based

10. **Ans (A,B,C,D)**

For option A, B

for maxima at O

$$d = n\lambda$$

For minima at O

$$d = (2n - 1) \frac{\lambda}{2}$$

For (C) At any point

$$I_R = I_{\max} \cos^2 \frac{\phi}{2}$$

$$\phi = \frac{2\pi}{\lambda} (\Delta x)$$

$$= \frac{2\pi}{\lambda} \left(\frac{11\lambda}{6} \right) = \frac{11\pi}{3}$$

$$\therefore I_R = \frac{3}{4} I_{\max}$$

For (D)

$$\Delta x = d \cos \theta = \frac{\lambda}{2}$$

$$\theta = 60^\circ$$

PART-1 : PHYSICS

SECTION-III

1. **Ans (1)**

$$R_V = 6\Omega$$

$$R_A = 0.5\Omega$$

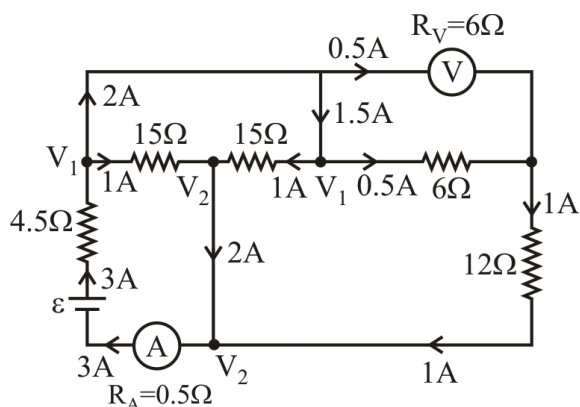
$$R_{eq} = 10\Omega$$

Reading of ammeter is

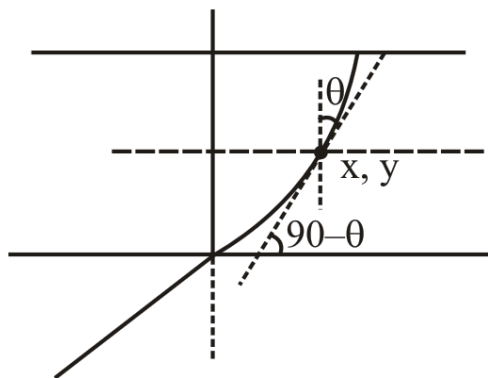
$$i = \frac{\varepsilon}{R_{eq}} = 3A$$

$$\text{reading of voltmeter} = 0.5 \times 6 = 3V$$

$$\text{ratio} = 1$$



2. **Ans (3)**



$$1 \sin 30 = \mu \sin \theta \quad \dots (i)$$

$$\& \tan(90 - \theta) = \frac{dy}{dx}$$

$$\cot \theta = 8x = 4\sqrt{y}$$

$$\text{at } y = \frac{1}{2}$$

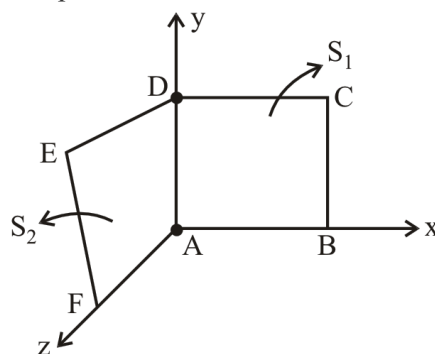
$$\cot \theta = 2\sqrt{2} \text{ therefore } \sin \theta = \frac{1}{3}$$

$\therefore$ from equation (i)

$$\mu = \frac{3}{2}$$

3. **Ans (2)**

Loop is as shown



$$\phi = \vec{B} \cdot (\vec{S}_1 + \vec{S}_2) = 2B_0L^2$$

$$\therefore n = 2$$

4. **Ans (9)**

Initially using equation of charging of capacitor

$$q = CE(1 - e^{-t/RC})$$

$$t = RC \ln 2$$

$$V_1 = \frac{q}{C} = \varepsilon(1 - e^{-\ln 2})$$

$$V_1 = \frac{\varepsilon}{2}$$

Now using LC oscillation.

$$q_i = CV_1 = \frac{C\varepsilon}{2}$$

$$\therefore q = q_i \cos\left(\frac{1}{\sqrt{LC}}t\right)$$

$$\therefore V_2 = \frac{q}{C} = V_1 \cos\left(\frac{1}{\sqrt{LC}}t\right)$$

$$= \frac{\varepsilon}{2} \cos\left(\frac{1}{\sqrt{LC}}t\right)$$

$$\text{Put } t = \frac{\pi}{3} \sqrt{LC}$$

$$V_2 = \frac{\varepsilon}{4}$$

$$\therefore V_1 + V_2 = \frac{3\varepsilon}{4} = 9V$$

5. **Ans (1)**

Position of maxima from O'

$$x_2 = \frac{2\lambda y}{dt}, x_1 = \frac{\lambda y}{d} \text{ (where } y \text{ is scree distance from O)}$$

differentiate w.r.t. time

$$x_2' = \frac{2\lambda y'}{dt}, x_1' = \frac{\lambda y'}{d}$$

$$\therefore x_2' - x_1' = \frac{\lambda y'}{d} = \frac{\lambda(gt)}{d} = 1 \text{ cm/s}$$

6. Ans (1)

$$\begin{aligned} \text{here, } C_{eq} &= \frac{2\epsilon_0 A}{d} \\ &= \frac{2 \times 8.852 \times 10^{-12} \times 1}{8.852 \times 10^{-4}} \\ &= 2 \times 10^{-8} \text{ F} \\ U &= \frac{1}{2} C_{eq} \epsilon^2 = 1 \times 10^{-6} = 1 \mu\text{J} \end{aligned}$$

7. Ans (2)

$$\frac{12}{X + Y + r} = 1 \dots \text{(i) (r is internal resistance of cell)}$$

$$5 = (1) X \dots \text{(ii)}$$

$$X = 5 \Omega$$

When Y is shortened current is

$$i = \frac{12}{5 + r}$$

Across battery

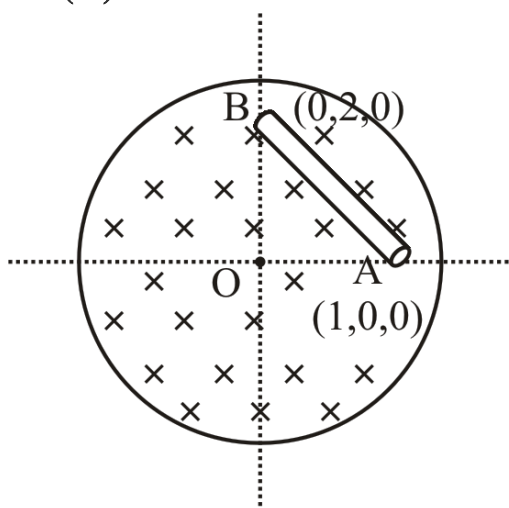
$$V = \epsilon - ir$$

$$8 = 12 - \frac{12}{5 + r} r$$

$$r = 2.5 \Omega$$

$$\therefore \frac{X}{r} = 2$$

8. Ans (2)

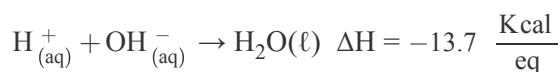


$$\begin{aligned} e &= \frac{d\phi}{dt} \\ &= \text{Area} \times \frac{dB}{dt} \end{aligned}$$

PART-2 : CHEMISTRY

SECTION-I (i)

2. Ans (A)



$$\text{no. of } m_{\text{eq}} \text{ of } \text{OH}^- = 224 \times 0.1 \times 1$$

$$= 22.4$$

$$\text{no. of } m_{\text{eq}} \text{ of } \text{H}^+ = 50 \times 0.1 \times 2$$

$$= 10 \text{ (Limiting reagent)}$$

An neutralisation 10 m_{eq} of acid as well as

10 m_{eq} of base react with other as we know that

$$1 \text{ eq} = 13.7 \text{ (Kcal)} \quad \text{(released)}$$

$$1000 m_{\text{eq}} = 13.7 \text{ Kcal (released)}$$

$$\text{So for } 10 m_{\text{eq}} = \frac{13.7}{1000} \text{ Kcal} \times 10 \text{ (released)}$$

$$= 137 \text{ cal (released)}$$

This release heat energy is gain by solution only

as heat capacity of calorimeter is negligible (given)

$$V_{(\text{solution})} = 224 + 50 = 274 \text{ ml}$$

$$Q/p = m_{\text{solution}} + S_{\text{solution}} \times \Delta T$$

$$= \rho_{\text{solution}} \times V_{\text{solution}} \times S_{\text{solution}} \times \Delta T$$

$$137 = 1 \times 274 \times 1 \times \Delta T$$

$$\Delta T = 0.5 \text{ K}$$

3. Ans (B)

$$\Delta S^\circ = S^\circ_{\text{CH}_4} - 2S^\circ_{\text{H}_2} - S^\circ_{\text{C}}$$

$$= 186.2 - 2 \times 130.6 - 6$$

$$= -81 \text{ J/K}$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = -T\Delta S_{\text{uni}}$$

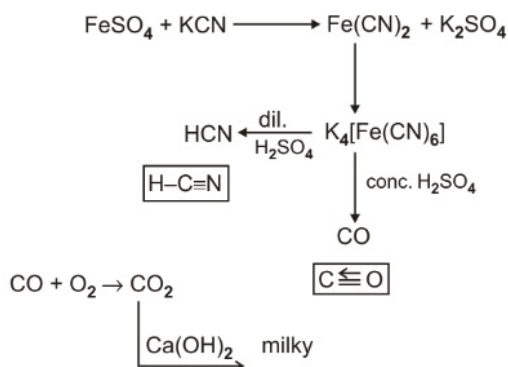
$$-75000 - 300(-81) = -300 \Delta S_{\text{uni}}$$

$$\therefore \Delta S_{\text{uni}} = 169$$

PART-2 : CHEMISTRY

SECTION-I (ii)

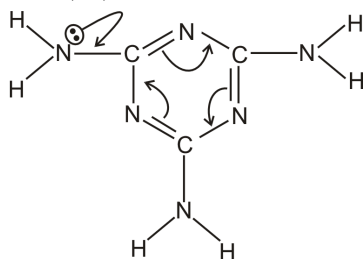
7. Ans (A,B,C)



PART-2 : CHEMISTRY

SECTION-III

1. Ans (6)



4. Ans (9)

Given : $\rightarrow$

1. $\rightarrow$ A and B are in equimolar solution.

$$X'_A = X'_B = 0.5$$

$$P'_T = 45 \text{ Torr}, P_A^0 = 20 \text{ Torr}$$

$$P'_T = X'_A P_A^0 + X'_B P'_B$$

$$45 = 0.5 \times 20 + 0.5 P_B^0$$

$$45 - 10 = 0.5 P_B^0$$

$$P_B^0 = 70 \text{ Torr}$$

2. $P_T = 22.5$ Torr

$$P_T = X_A P_A^0 + X_B P_B^0, X_A + X_B = 1 \quad X_A = 1 - X_B$$

$$P_T = (1 - X_B) P_A^0 + X_B P_B^0$$

$$22.5 = 20 - 20X_B + 70 X_B$$

2.5 = 50 XB

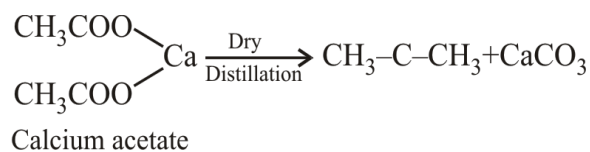
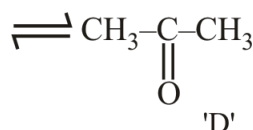
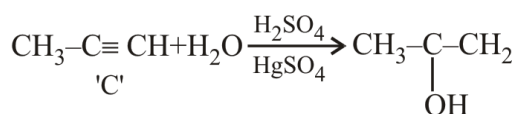
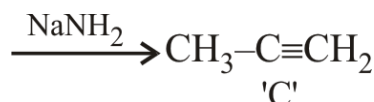
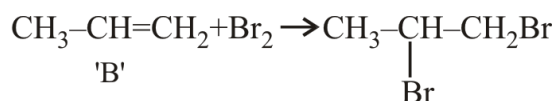
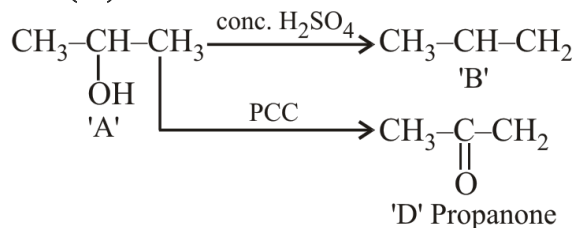
$$X_B = 0.05$$

$$X_A = 1 - 0.05 = 0.95$$

$$y = \frac{X_A}{X_B} = \frac{0.95}{0.05} = 19$$

$$y - 10 = 19 - 10 = \boxed{9}$$

5. Ans (4)



The molecular weight of 'C'

(C₃H₄) is 36 + 4 = 40 g mol⁻¹

$$= 4 \times 10 \text{ hence } Y = 4$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (B)**

For $1 \leq k \leq n$

$$\left(1 + \frac{x}{n}\right)^{k/n} \leq \left(1 + \frac{kx}{n^2}\right) \leq \left(1 + \frac{x}{n^2}\right)^k$$

$$\left(1 + \frac{x}{n}\right)^{\frac{n+1}{2}} \leq P_n \leq \left(1 + \frac{x}{n^2}\right)^{\frac{n(n+1)}{2}}$$

$$e^{x/2} \leq \lim_{n \rightarrow \infty} P_n \leq e^{x/2} \Rightarrow f(x) = e^{x/2}$$

For $0 < x < \infty$

$$e^{x/2} < e^x$$

$$\Rightarrow f(x) < e^x$$

2. Ans (D)

$$f(x+y) - xy = f(x) + f(y)$$

$$\text{put } y = 1$$

$$\Rightarrow f(x+1) - x = f(x) + f(1)$$

$$\Rightarrow f(x+1) - f(x) = x + f(1)$$

$$\text{put } x = 1, 2, 3, \dots, x$$

$$f(2) - f(1) = 1 + f(1)$$

$$f(3) - f(2) = 2 + f(1)$$

$$f(4) - f(3) = 3 + f(1)$$

⋮

$$f(x+1) - f(x) = x + f(1)$$

$$f(x+1) - f(1) = \frac{x(x+1)}{2} + xf(1)$$

$$f(x+1) = \frac{x(x+1)}{2} + (x+1)f(1)$$

$$\text{put } x = 6, f(7) = \frac{6 \times 7}{2} + 7f(1)$$

$$\Rightarrow f(1) = 0$$

$$\text{so, } f(x+1) = \frac{x(x+1)}{2}$$

$$\Rightarrow f(x) = \frac{(x-1)x}{2}$$

$f^{-1}(x)$ will not exist

3. Ans (D)

$$\int \frac{x^3 - x^2 + x - 1}{x^5 + 1} dx = \int \frac{x^4 - 1}{(x^5 + 1)(x + 1)} dx$$

$$= \int \frac{x^4}{x^5 + 1} dx - \int \frac{dx}{(x + 1)}$$

$$= \frac{1}{5} \int \frac{5x^4}{x^5 + 1} dx - \ln|x + 1|$$

$$= \frac{1}{5} \ln|1 + x^5| - \ln|x + 1| + C$$

$$= \frac{1}{5} \ln \left| \frac{1 + x^5}{(1 + x)^5} \right| + C$$

$$\Rightarrow f(x) = \frac{1}{5} \ln \left| \frac{1 + x^5}{(1 + x)^5} \right| + C$$

$$f(0) = C = 0$$

$$\Rightarrow f(x) = \frac{1}{5} \ln \left| \frac{1 + x^5}{(1 + x)^5} \right|$$

4. Ans (B)

$f(x)$ is continuous at $x = 0$

$$\Rightarrow \lim_{x \rightarrow 0} x^n \cdot \sin\left(\frac{1}{x^3}\right) = 0$$

$$\Rightarrow 0 < n$$

$f(x)$ is not differentiable at $x = 0$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{f(0+h)}{h} \text{ must not exist}$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{h^n \sin\left(\frac{1}{h^3}\right)}{h} \text{ must not exist}$$

$$\Rightarrow n - 1 \leq 0 \Rightarrow n \leq 1$$

$$\text{so } 0 < n \leq 1$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,B,C,D)

$$\int_1^n \frac{\sin x}{x^2} dx < \int_1^n \frac{x}{x^2} dx$$

$$= \ln x \Big|_1^n = \ln(n)$$

$$\Rightarrow I(n) < \ln(n)$$

$$\Rightarrow I(2) < \ln(2)$$

$$\Rightarrow I(3) < \ln(3)$$

$$\text{Also } \sin x = x - \frac{x^3}{3!} \dots > x - \frac{x^3}{6}$$

$$\Rightarrow \frac{\sin x}{x^2} > \frac{1}{x} - \frac{x}{6}$$

$$\Rightarrow \int_1^n \frac{\sin x}{x^2} dx > \ln n - \left(\frac{n^2 - 1}{12}\right)$$

$$\Rightarrow I(2) > \ln 2 - \frac{1}{4}$$

For $1 < x < 2$

$$\sin x > \frac{1}{\sqrt{2}}$$

$$\Rightarrow \frac{\sin x}{x^2} > \frac{1}{\sqrt{2}x^2}$$

$$\Rightarrow \int_1^2 \frac{\sin x}{x^2} dx > \int_1^2 \frac{1}{\sqrt{2}x^2} dx = \frac{1}{2\sqrt{2}}$$

6. Ans (C)

Line of intersection of $P_1 = 0$ & $P_2 = 0$

line of intersection is $\frac{x}{0} = \frac{y-4}{-4} = \frac{z}{5}$

$$(A) \frac{x-1}{0} = \frac{y-1}{0} = \frac{z-1}{5}$$

is skew pair with L.O.I.

so, it can be an edge.

$$(B) \frac{x-6}{-5} = \frac{y}{2} = \frac{z}{3}$$

is skew pair with L.O.I.

so, it can be an edge

$$(C) \frac{x}{-2} = \frac{y-4}{5} = \frac{z}{4}$$

is intersecting L.O.I.

but not lying in any plane

$P_1 = 0$ or $P_2 = 0$

so, it cannot be an edge.

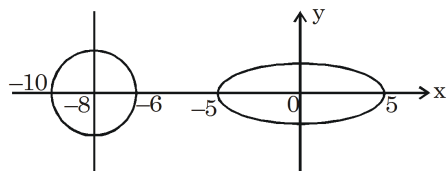
$$(D) \frac{x}{1} = \frac{y-4}{-2} = \frac{z}{3}$$

is intersecting L.O.I.

and lying in $P_2 = 0$

so, it can be an edge.

7. Ans (B,C)



$$|z_1 - z_2| \in [1, 15]$$

8. Ans (A,B,C,D)

$$P(A \cap B) = P(A) \cdot P(B) = \frac{1}{12}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{1}{3} + \frac{1}{4} - \frac{1}{12} = \frac{1}{2}$$

$$P(B \cup \bar{A}) = P(B) + P(\bar{A}) - P(B) \cdot P(\bar{A})$$

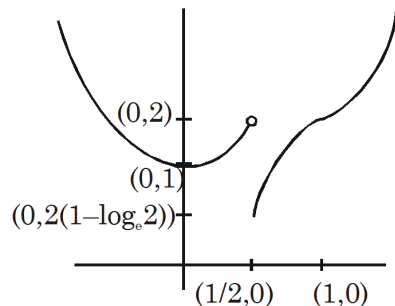
$$= \frac{1}{4} + \frac{2}{3} - \frac{1}{4} \times \frac{2}{3} = \frac{3}{4}$$

$$P(A \cup \bar{B}) = P(A) + P(\bar{B}) - P(A) \cdot P(\bar{B})$$

$$= \frac{1}{3} + \frac{3}{4} - \frac{1}{3} \cdot \frac{3}{4} = \frac{5}{6}$$

9. Ans (A,B,C)

$$f(x) = \begin{cases} x^2 + 1 & x \geq 1 \\ 2(\log_e x + 1) & \frac{1}{2} \leq x < 1 \\ 4x^2 + 1 & x < \frac{1}{2} \end{cases}$$

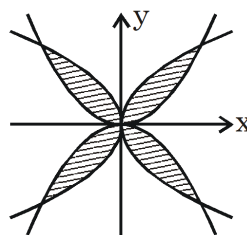


$f(x)$ is non-differentiable at 1 point

$f(x)$ is discontinuous at 1 point

$f(x)$ has 2 points of minima

10. Ans (A,C)



$$\text{Area} = 4 \times \frac{16ab}{3}$$

$$= 4 \times \left(\frac{16 \times 2 \times \frac{3}{4}}{3} \right) = 32$$

PART-3 : MATHEMATICS

SECTION-III

1. Ans (1)

$$f'(x) + f(x) \cdot \frac{x}{1+x^2} = \frac{x}{1+x^2}$$

$$\text{If } f = \sqrt{1+x^2}$$

$$\therefore \sqrt{1+x^2} f(x) = \int \frac{x}{\sqrt{1+x^2}} dx + c$$

$$= \sqrt{1+x^2} + c$$

$$\therefore f(x) = 1 + \frac{c}{\sqrt{1+x^2}}$$

$$\therefore f(0) = \frac{4}{3} \therefore c = \frac{1}{3}$$

$$\therefore f(\sqrt{8}) = 1 + \frac{1}{9} = \frac{10}{9}$$

$$\therefore f(\sqrt{8}) - \frac{1}{9} = 1$$

2. Ans (3)

$$\begin{aligned} |A| &\neq 0 \\ |A^{2020}(A^2 - (\beta - 1)A - \beta I)| &= 0 \\ \Rightarrow |A^{2020}(A^2 + A - \beta(A + I))| &= 0 \\ \Rightarrow |A^{2020}(A(A + I) - \beta(A + I))| &= 0 \\ \Rightarrow |A^{2020}(A - \beta I)(A + I)| &= 0 \\ |A^{2020}| \neq 0, |A + I| &\neq 0 \\ \Rightarrow |A - \beta I| &= 0 \Rightarrow \beta = \frac{1}{3} \end{aligned}$$

3. Ans (7)

$$\begin{aligned} \vec{r} &= \hat{k} + t(-\hat{i} + \hat{j}) + p(-\hat{i} + \hat{k}) \\ \vec{n} &= \hat{i} + \hat{j} + \hat{k} \\ \Rightarrow x + y + z &= 1 \\ Q(10, 15, 20) \text{ and } (\alpha, \beta, \gamma) \\ \frac{\alpha - 10}{1} &= \frac{\beta - 15}{1} = \frac{\gamma - 20}{1} = -2 \left(\frac{10 + 15 + 20 - 1}{1 + 1 + 1} \right) \\ &= -\frac{88}{3} \\ \Rightarrow (\alpha, \beta, \gamma) &\equiv \left(-\frac{58}{3}, -\frac{43}{3}, -\frac{28}{3} \right) \\ |\alpha| + |\beta| + |\gamma| &= \frac{58 + 43 + 28}{3} \\ &= \frac{129}{3} = 43 \end{aligned}$$

4. Ans (4)

Given that

$$z \neq \bar{z}$$

Let

$$\begin{aligned} \alpha &= \frac{2 + 3z + 4z^2}{2 - 3z + 4z^2} = \frac{(2 - 3z + 4z^2) + 6z}{2 - 3z + 4z^2} \\ \therefore \alpha &= 1 + \frac{6z}{2 - 3z + 4z^2} \end{aligned}$$

If α is a real number, then

$$\alpha = \bar{\alpha}$$

$$\Rightarrow \frac{z}{2 - 3z + 4z^2} = \frac{\bar{z}}{2 - 3\bar{z} + 4\bar{z}^2}$$

$$\therefore 2(z - \bar{z}) = 4z\bar{z}(z - \bar{z})$$

$$\Rightarrow (z - \bar{z})(2 - 4z\bar{z}) = 0$$

As $z \neq \bar{z}$ (Given)

$$\Rightarrow z\bar{z} = \frac{2}{4} = \frac{1}{2}$$

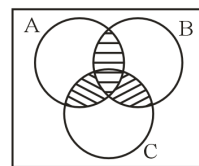
$$\Rightarrow |z|^2 = \frac{1}{2}$$

$$\Rightarrow 8|z|^2 = 4$$

5. Ans (2)

$$n(U) = 900$$

Let $A \equiv$ Fever, $B \equiv$ Cough



$C \equiv$ Breathing problem

$$\therefore n(A) = 190, n(B) = 220, n(C) = 220$$

$$n(A \cup B) = 330, n(B \cup C) = 350,$$

$$n(A \cup C) = 340, n(A \cap B \cap C) = 30$$

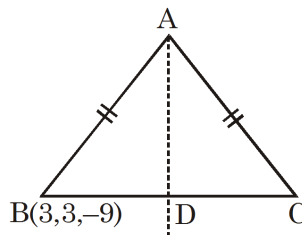
$$\Rightarrow n(A \cap B) = 80, n(B \cap C) = 90, n(C \cap A) = 70$$

$$\text{Favourable no} = 80 + 90 + 70 - 60 = 180$$

$$p = \frac{180}{900} \Rightarrow 10p = 2$$

6. Ans (9)

Let D is $(\lambda + 1, 2\lambda + 2, 3\lambda - 3)$



$$\vec{BD} \cdot (\hat{i} + 2\hat{j} + 3\hat{k}) = 0 \Rightarrow \lambda = -1$$

$\therefore D$ is mid point of BC

$\therefore C$ is $(-3, -3, -3)$

7. Ans (2)

$$y - x \frac{dy}{dx} = x^3 \Rightarrow xdy - ydz = -x^3 dx$$

$$\Rightarrow \frac{xdy - ydx}{x^2} = -x dx$$

$$\Rightarrow \frac{y}{x} = -\frac{x^2}{2} + c$$

$$\frac{1}{2} = -\frac{1}{2} + c = 0 \Rightarrow c = 1$$

$$\Rightarrow y = -\frac{x^3}{2} + x$$

8. Ans (1)

$$= (\cos x)^{2018} \sin(2019x + x)$$

$$= (\cos x)^{2018} (\sin(2019x)\cos x + \cos(2019x)\sin x)$$

$$= - \int_0^{\pi/2} \frac{d}{dx} \left(\cos(2019x)(\cos x)^{2019} \right) dx$$

$$= |-1| = 1$$